📊 RURO Post-Estimation Analysis

Generated: 2026-05-15 10:31:29

Total Parameters: 47 | **Groups:** joint

Estimation Results Source:

`U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0c_b2\run_2026-05-15_10-05-45\estimation_results.json`

Estimation Configuration

Specification	ruro_occ_M0c_b2
Specification File	`U:\Desktop\Nizam_Hisham\MNL\scripts\enhanced\estimation_spec_ruro_occ_M0c_b2.yaml`
Model Family	regular
Proposal Correction	Enabled
Proposal Correction Form	-log(prior)
Opportunity Centering	Enabled

Single Males Single Females Males in Couples Females in Couples

⏱️ Elapsed Time

Estimation	Post-Estimation	Total	Iterations
4m 16s	21.2s	4m 37s	20

📈 Model Fit Statistics

log_likelihood	-6509.1602
n_observations	4.2530e+05

n_groups	4253.0000
n_parameters	47.0000
AIC	1.3112e+04
BIC	1.3627e+04
ll_null	-1.9586e+04
ll_null_uniform	-1.9586e+04
ll_null_prior_corrected	-2.2322e+04
n_obs_long	4.2530e+05
rho_squared_uniform	0.6677
rho_squared_adj_uniform	0.6653
rho_squared_prior_corrected	0.7084
rho_squared_adj_prior_corrected	0.7063
rho_squared	0.7084
rho_squared_adj	0.7063
AIC_per_obs	0.0308
n_negative_muc_total	0.0000e+00
n_negative_mul_total	0.0000e+00
pct_negative_muc_total	0.0000e+00
pct_negative_mul_total	0.0000e+00
n_bounded_params	47
n_hit_lower_bound	0
n_hit_upper_bound	0

📝 Understanding Bounded Parameters

- **n_bounded_params (47)**: Parameters with constraints defined.
- **n_hit_lower_bound (0)**: Parameters at lower bound. ✓ Good!

- **n_hit_upper_bound (0)**: Parameters at upper bound. ✓ Good!

🔍 Identification Diagnostics

Hessian-based diagnostics for parameter identification quality.

Condition Number: 5.06e+10

✗ Severe Identification Problem

The Hessian is severely ill-conditioned (κ ≥ 10,000). This indicates serious identification issues with unreliable parameter estimates.

Eigenvalue Analysis

Minimum Eigenvalue

-1.39e+01

Maximum Eigenvalue

1.37e+10

Negative Eigenvalues

1

✗ NOT a local maximum!

Eigenvalue Summary

p5 / median / p95: 1.67e+00 / 1.97e+02 / 8.08e+05

Smallest 5: -1.39e+01, 2.70e-01, 1.59e+00, 1.86e+00, 3.21e+00

Largest 5: 4.97e+05, 5.77e+05, 9.07e+05, 9.89e+05, 1.37e+10

High Correlations (|rho| >= 0.90)

Parameter A	Parameter B	Correlation
`theta_c_singles`	`beta_c`	-1.084
`beta_c_sm`	`beta_c_sf`	-1.054
`beta_c_sf`	`theta_c_singles`	-1.044

Parameter A	Parameter B	Correlation
beta_c_sm	theta_c_singles	-1.034
beta_w_pexp	beta_w_pexp2	-0.960
beta_E	beta_E_gsur	-0.950
beta_c_sm	beta_c	-0.950
beta_c_sf	beta_c	-0.929

ℹ️ Technical Notes

Condition Number (κ): Measures sensitivity of parameter estimates to small changes in data. $\kappa = \lambda_{max} / \lambda_{min}$, where λ are eigenvalues of -H (Hessian).

Eigenvalues: All eigenvalues of -H should be positive for a local maximum. Near-zero eigenvalues indicate flat directions (weak identification).

Sign convention: Eigenvalues shown come from the Hessian used to compute SEs. When SEs are computed numerically, this is the Hessian of the negative log-likelihood (H), so negative values indicate H is not positive semidefinite (ill-conditioning or non-optimum).

Standard Errors: Computed as $SE = \sqrt{\mathrm{diag}((-H)^{-1})}$, where H is the Hessian at the optimum. Large SE indicates parameter is not precisely estimated.

🧪 Identification Diagnostics (Estimation Stage)

Source:

U:\Desktop\Nizam_Hisham\MNL\outputs\estimates\fr\spec\ruro_occ\gamspy\estimation_spec_ruro_occ_M0c_b2\run_2026-05-15_10-05-45\identification_diagnostics.txt

```
Identification Diagnostics
================================================================================
specification: ruro_occ_M0c_b2
n_parameters: 47
bounded_hits_total: 0
bounded_hits_lower: 0
bounded_hits_upper: 0
hessian_condition_number: 50595697514.8011
hessian_negative_eigenvalues: 1
```

```
hessian_near_zero_eigenvalues(|eig|<=1e-8): 0
negative_variances_from_varcov: 3

Parameter Stability
--------------------------------------------------------------------------------
reference: initial_vector
matched_params: 47/47
delta_l2: 7.823210e+00
delta_l2_relative: 1.780741e+00
delta_max_abs: 3.458840e+00
delta_mean_abs: 6.688398e-01

Bound Hits
--------------------------------------------------------------------------------
none
```

📐 Model Specification

Utility and opportunity functions for each demographic group, shown in both symbolic and numerical form.

Single Males

📐 Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^{\theta} - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2$

🔢 Utility Function (Numerical)

U = 0.6358 · BC(C, 0 (log)) + (3.8740 + 0.0085 · age + 0.0020 · age2) · BC(L, -0.7119)

Single Females

📐 Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^{\theta} - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,nkids} \cdot nkids$

🔢 Utility Function (Numerical)

```
U = 0.5760 · BC(C, 0 (log)) + (4.4588 + 0.0019 · age + 0.0041 · age2 +
0.0567 · nkids) · BC(L, -0.7280)
```

Males in Couples

📐 Utility Function (Symbolic)

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^{\theta} - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2$

🔢 Utility Function (Numerical)

```
U = 4.0515 · BC(C, 0 (log)) + (0.0119 - 0.0079 · age + 0.0006 · age2) ·
BC(L, -0.7319)
```

Females in Couples

📐 **Utility Function (Symbolic)**

$$U = \beta_c \cdot BC(C, \theta_c) + \beta_l(X) \cdot BC(L, \theta_l)$$

where $BC(x, \theta) = (x^\theta - 1) / \theta$ (Box-Cox transformation)
$\beta_l(X) = \beta_{l0} + \beta_{l,age} \cdot age + \beta_{l,age2} \cdot age2 + \beta_{l,nkids} \cdot nkids$

🔢 **Utility Function (Numerical)**

$$U = 4.0515 \cdot BC(C, 0\ (\log)) + (2.5821 - 0.0574 \cdot age + 0.0028 \cdot age2 + 0.1769 \cdot nkids) \cdot BC(L, -0.6777)$$

📐 Model Index

$$V_{ij} = U_{ij} + O^E_{ij} + O^H_{ij} + O^W_{ij} + O^{Occ}_{ij} + - \log prior_{ij}$$

$$P_{ij} = \exp(V_{ij}) / \Sigma_{k\in C_i} \exp(V_{ik})$$

U is the utility of consumption and leisure; the O˙ terms are opportunity-density contributions (employment/hours, wage, occupation); `log prior` rescales each alternative to the data-generating measure. The terms displayed reflect the opportunity blocks declared in the active YAML specification.

⏰ Employment and Hours Opportunity Parameters

Working-gated employment and hours intercepts. Combines `hours_opportunity.shifters` (base employment intercept and PT/FT focal-point indicators) with `market_opportunity.shifters` (working-gated residual market access terms like gsur, education).

Symbolic form

```
                  O^E + O^H =

                      beta_E * working
    + beta_h_pt1 * working_pt1
    + beta_h_pt2 * working_pt2
    + beta_h_ft * working_ft
    + beta_E_gsur * gsur * working
    + beta_E_educH * educH * working
```

Numerical form (by group)

```
Joint (All Groups): -2.8423 * working  +  -0.4987 * working_pt1  +
0.3650 * working_pt2  +  1.4441 * working_ft  +  -0.7438 * gsur *
working  +  0.6134 * educH * working
```

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: `log(wage) = μw(X) + ε, ε ~ N(0, σ²)`. The mean μ_W is built from `wage_opportunity.mean_shifters`.

Symbolic form

$$\mu_w =$$

```
            beta_w0
+ beta_w_educL · educL
+ beta_w_educH · educH
+ beta_w_pexp · pexp_years
+ beta_w_pexp2 · pexp_years2
```

$$\log(wage) = \mu_w + \varepsilon, \quad \varepsilon \sim N(0, sigma^2)$$

Numerical form (by group)

Joint (All Groups): μ_w = 2.0249 + -0.0510 * educL + 0.3161 * educH + 0.0181 * pexp_years + -0.0002 * pexp_years2

sigma = 0.4268

Occupation Opportunity Parameters

Working-gated occupation shifters. Each `applies_to` group has its own coefficient set on the non-reference categories of `loc4`. Reference category: `loc4 = 1`.

Single Males (`applies_to = sm`)

$$O^{Occ}_{sm} =$$

```
            beta_occ_2_sm * loc4_2 * working
+ beta_occ_3_sm * loc4_3 * working
+ beta_occ_4_sm * loc4_4 * working
```

```
-1.5104 * loc4_2 * working  +  -2.1651 * loc4_3 *
working  +  0.0237 * loc4_4 * working
```

Single Females (`applies_to = sf`)

$$O^{Occ}_{sf} =$$

```
        beta_occ_2_sf * loc4_2 * working
  + beta_occ_3_sf * loc4_3 * working
  + beta_occ_4_sf * loc4_4 * working
```

```
-0.0105 * loc4_2 * working  +  -0.5610 * loc4_3 *
working  +  0.7988 * loc4_4 * working
```

couples_male (`applies_to = cm`)

$$O^{Occ}_{cm} =$$

```
        beta_occ_2_cm * loc4_2 * working
  + beta_occ_3_cm * loc4_3 * working
  + beta_occ_4_cm * loc4_4 * working
```

```
-1.4760 * loc4_2 * working  +  -2.2240 * loc4_3 *
working  +  0.4725 * loc4_4 * working
```

couples_female (`applies_to = cf`)

```
O^Occ_cf =

        beta_occ_2_cf * loc4_2 * working
  + beta_occ_3_cf * loc4_3 * working
  + beta_occ_4_cf * loc4_4 * working
```

```
        0.1763 * loc4_2 * working  +  -0.2164 * loc4_3 *
working  +  1.1147 * loc4_4 * working
```

📈 Curvature-Based Heuristics (Structural Elasticity Approximations)

⚠️ Interpretation Note

These are **not** true labor supply elasticities. They are heuristic approximations derived from the curvature parameters (θ) of the Box-Cox utility function. The Hicksian approximation is (1 - θ_l), which measures the curvature of preferences. For rigorous elasticity estimates, use simulation-based methods that account for the full discrete choice structure and budget constraints.

📊 Labor Supply Elasticities

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Males	1.712	1.612	0.514	1.198	-0.712	0.500	3.874	0.636

Group	Hicksian (compensated)	Marshallian (uncompensated)	Participation (extensive)	Intensive (conditional)	θ_l	θ_c	β_l (at median X)	β_c
Single Females	1.728	1.628	0.518	1.210	-0.728	0.500	4.459	0.576
Males in Couples	1.732	1.632	0.520	1.212	-0.732	0.500	0.012	4.052
Females in Couples	1.678	1.578	0.503	1.174	-0.678	0.500	2.582	4.052

🎯 Fit Diagnostics

Observed vs Predicted

Group	Obs. Participation	Pred. Participation	Obs. Mean Hours	Pred. Mean Hours
Single Males	93.0%	90.8%	39.3	35.7
Single Females	94.0%	95.2%	36.3	35.1
Males in Couples	97.2%	98.3%	41.6	42.8
Females in Couples	96.5%	98.9%	35.6	39.0

⚠️ Marginal Utility Diagnostics

Analysis at chosen alternatives. Negative values indicate potential specification issues.

Negative MUC/MUL Summary

MUC Behavior Analysis

For well-behaved utility: MUC > 0 ($\beta_c > 0$) and diminishing ($\theta_c < 1$)

Group	N Individuals	% Neg MUC	% Neg MUL	MUC Mean	MUL Mean
Single Males	910	0.00%	0.00%	0.0147	7.3221e-03
Single Females	766	0.00%	0.00%	0.0134	9.8020e-03
Males in Couples	2577	0.00%	0.00%	0.0640	2.8324e-05
Females in Couples	2577	0.00%	0.00%	0.0640	5.4519e-03

Group	β_c	θ_c	MUC Positive?	MUC Diminishing?	Well-Behaved?	MUC at Median C	C where MUC=1	Notes
Single Males	0.6358	0.5000	✓	✓	✓	0.6358	0.4043	
Single Females	0.5760	0.5000	✓	✓	✓	0.5760	0.3318	
Males in Couples	4.0515	0.0000	✓	✓	✓	4.0515	4.0515	
Females in Couples	4.0515	0.0000	✓	✓	✓	4.0515	4.0515	

📊 Choice Probability Diagnostics

Probability Sum Sanity Check

Max |Σp - 1|	0.000000
Mean |Σp - 1|	0.000000
% HH off by >0.01	0.00%
% HH off by >0.001	0.00%

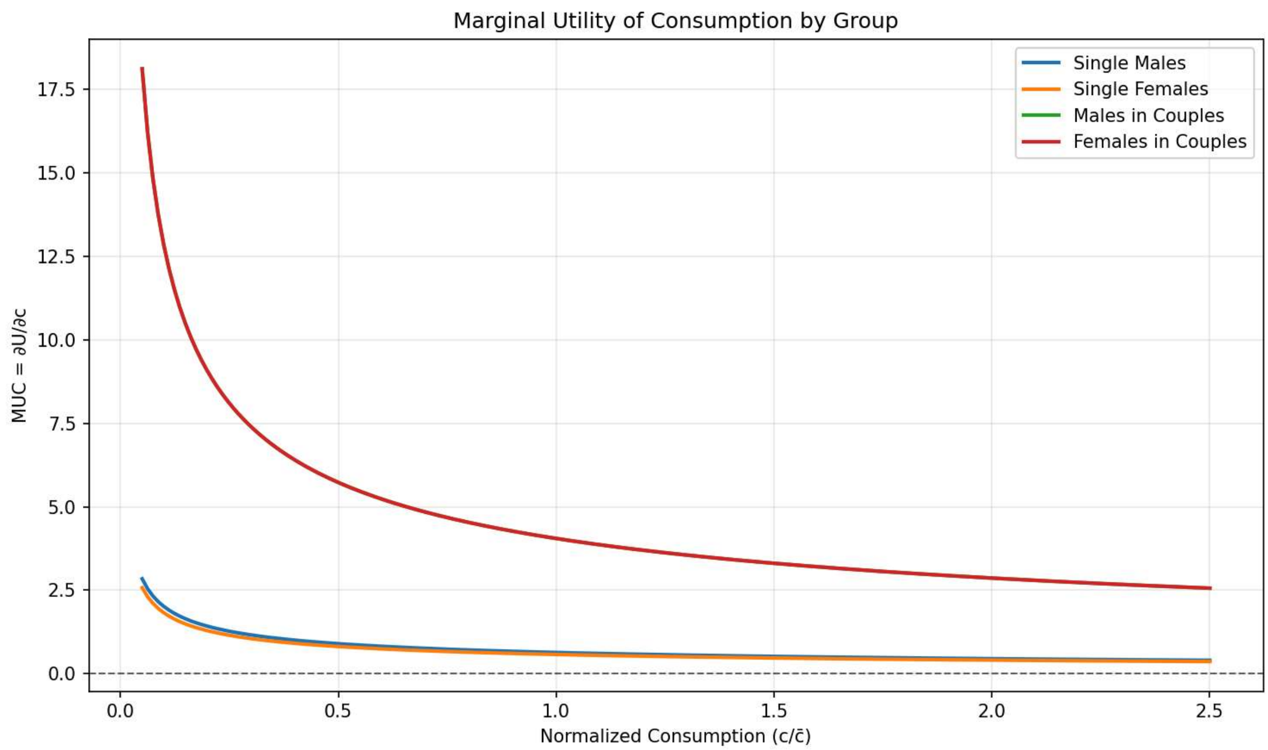
P(chosen) Distribution

Min	0.0000
10th percentile	0.0157
25th percentile	0.0671
Median	0.2696
Mean	0.3885
75th percentile	0.7229
90th percentile	0.9201
Max	0.9993

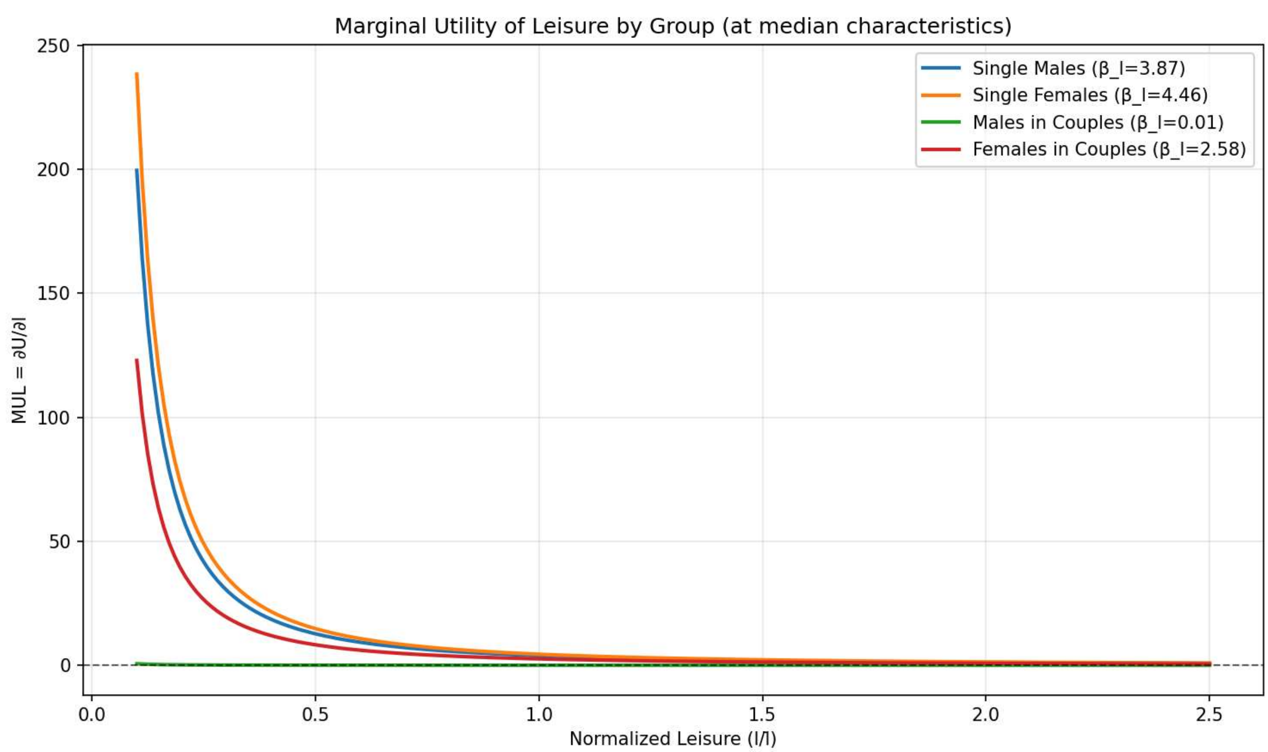
🗺️ Utility Contours & Plots

Marginal Utility Comparison

Marginal Utility of Consumption

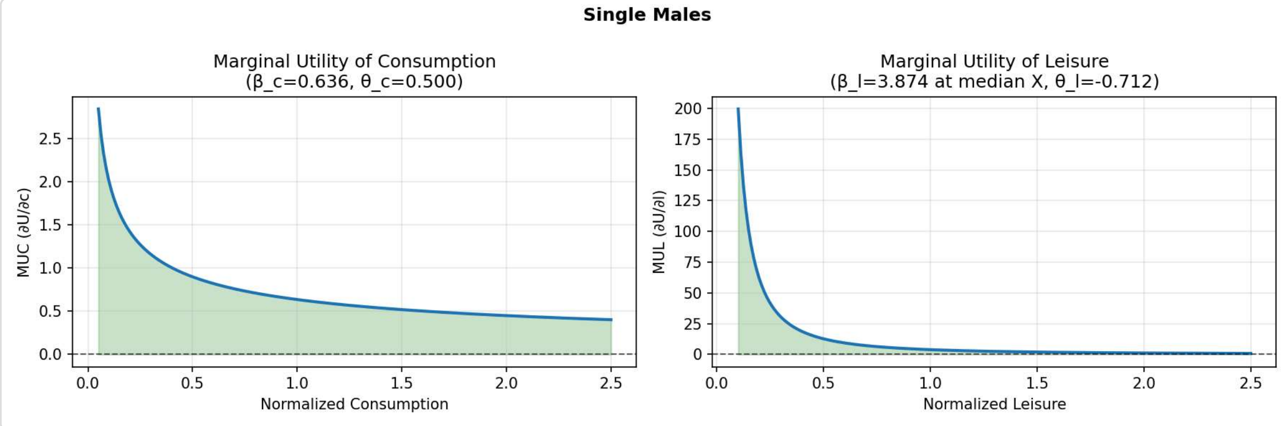


Marginal Utility of Leisure

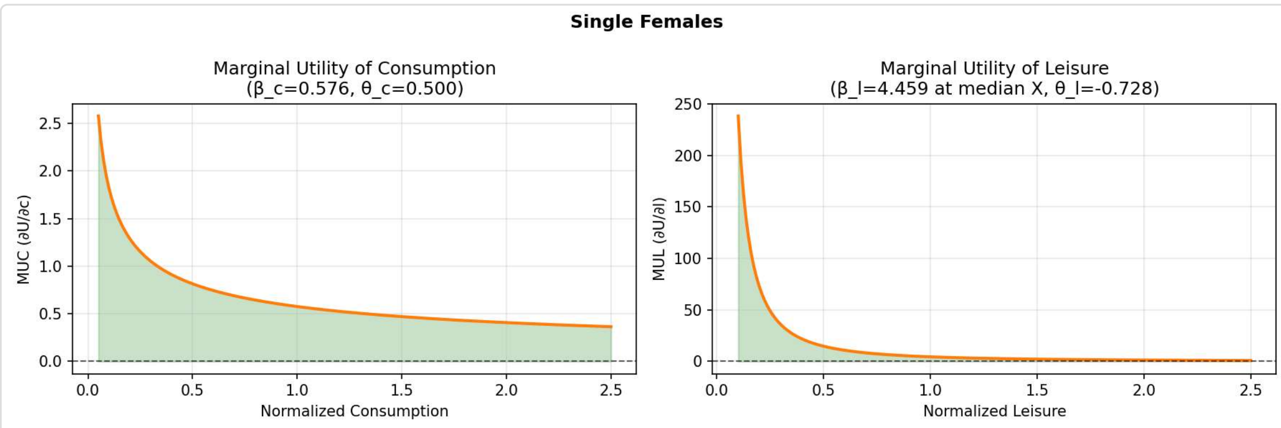


📊 Marginal Utility Distributions by Group

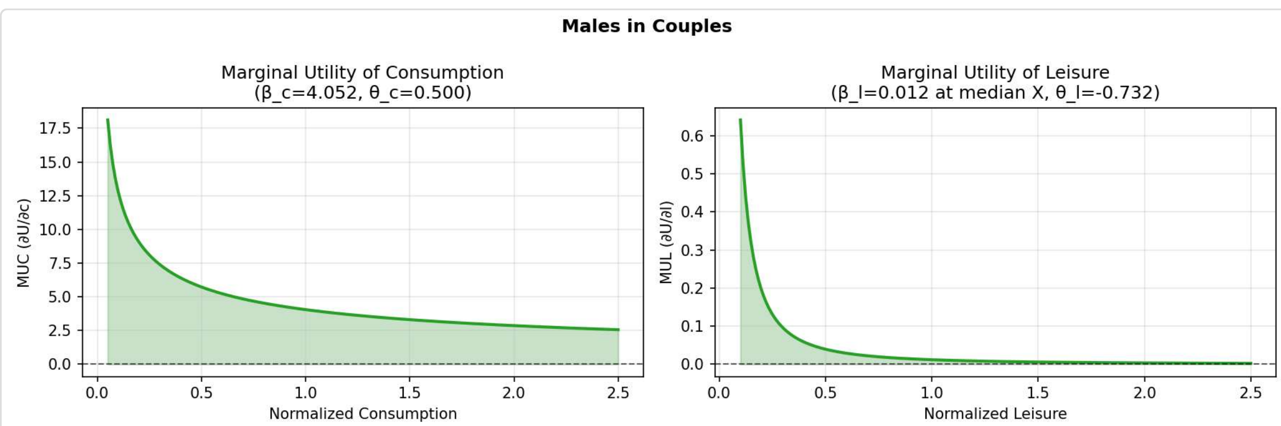
Single Males - MUC & MUL Distributions



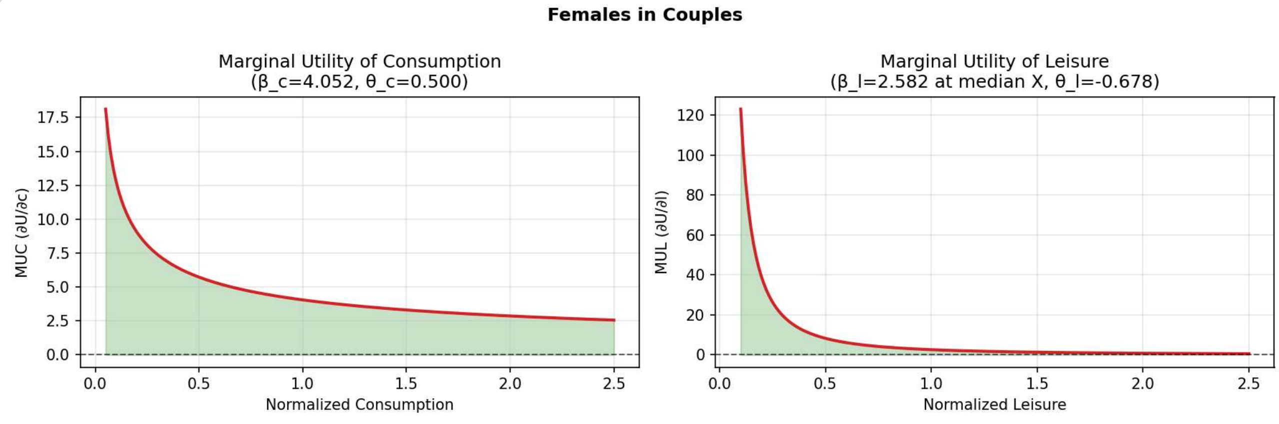
Single Females - MUC & MUL Distributions



Males in Couples - MUC & MUL Distributions

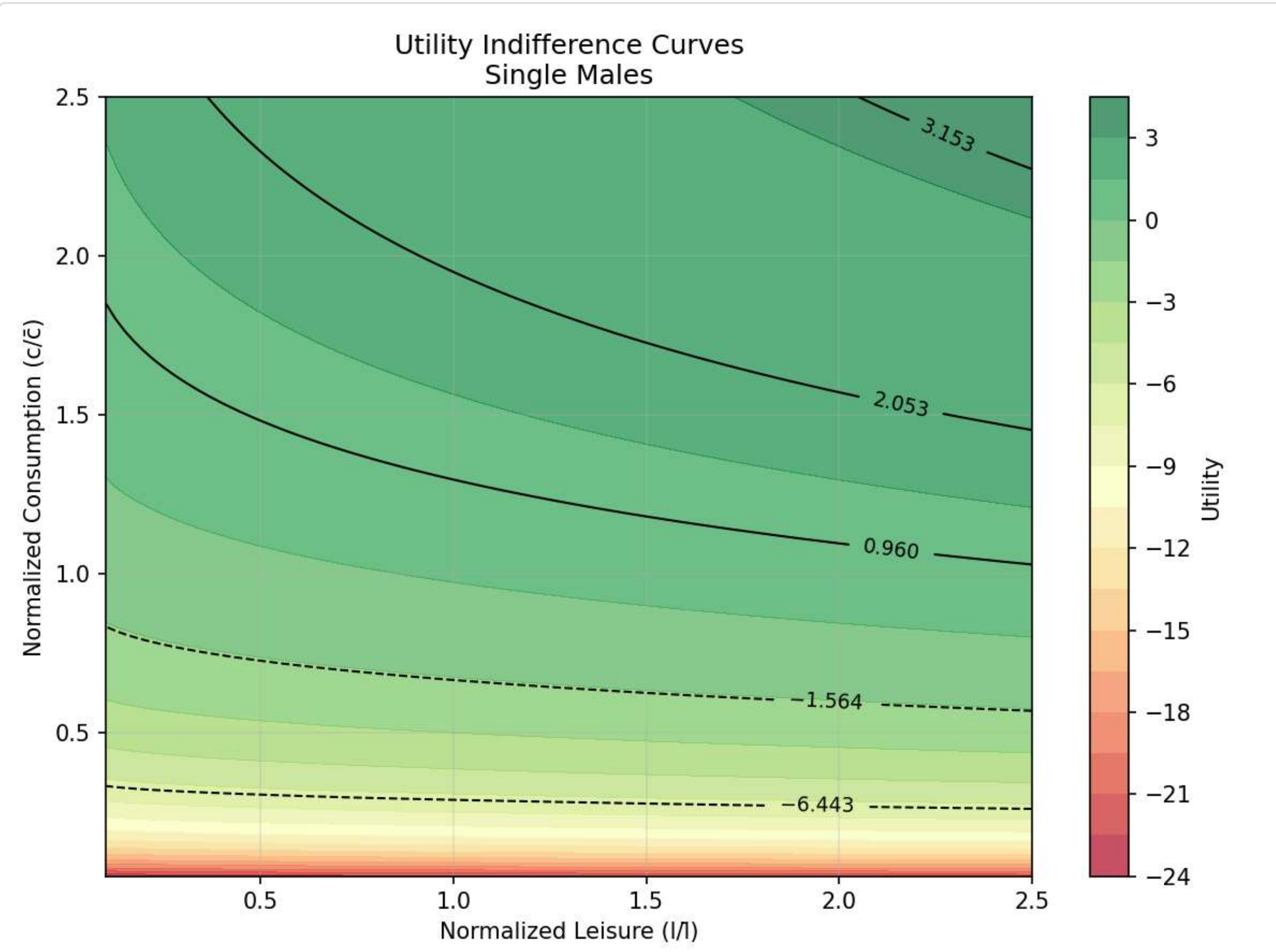


Females in Couples - MUC & MUL Distributions

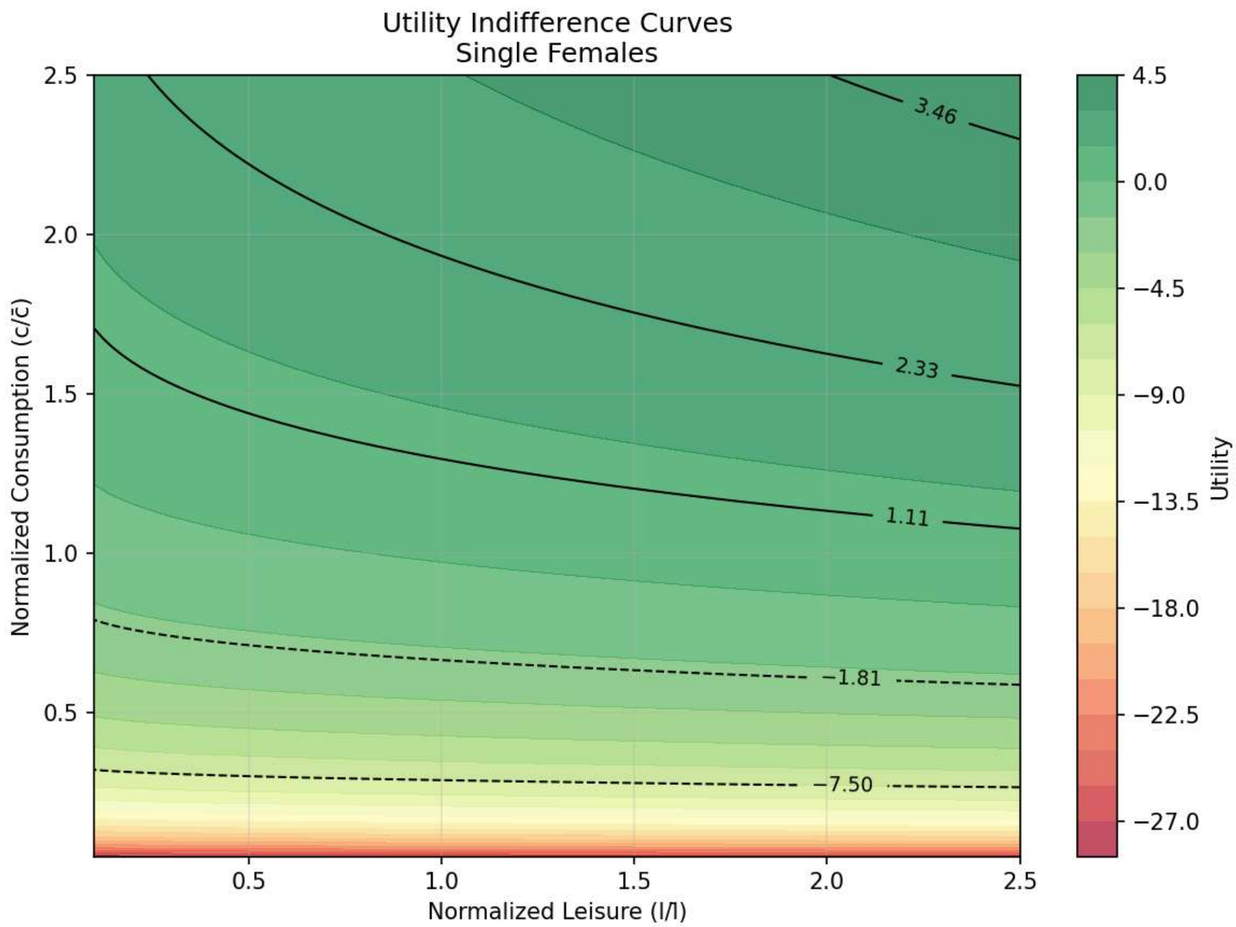


Utility Indifference Curves by Group

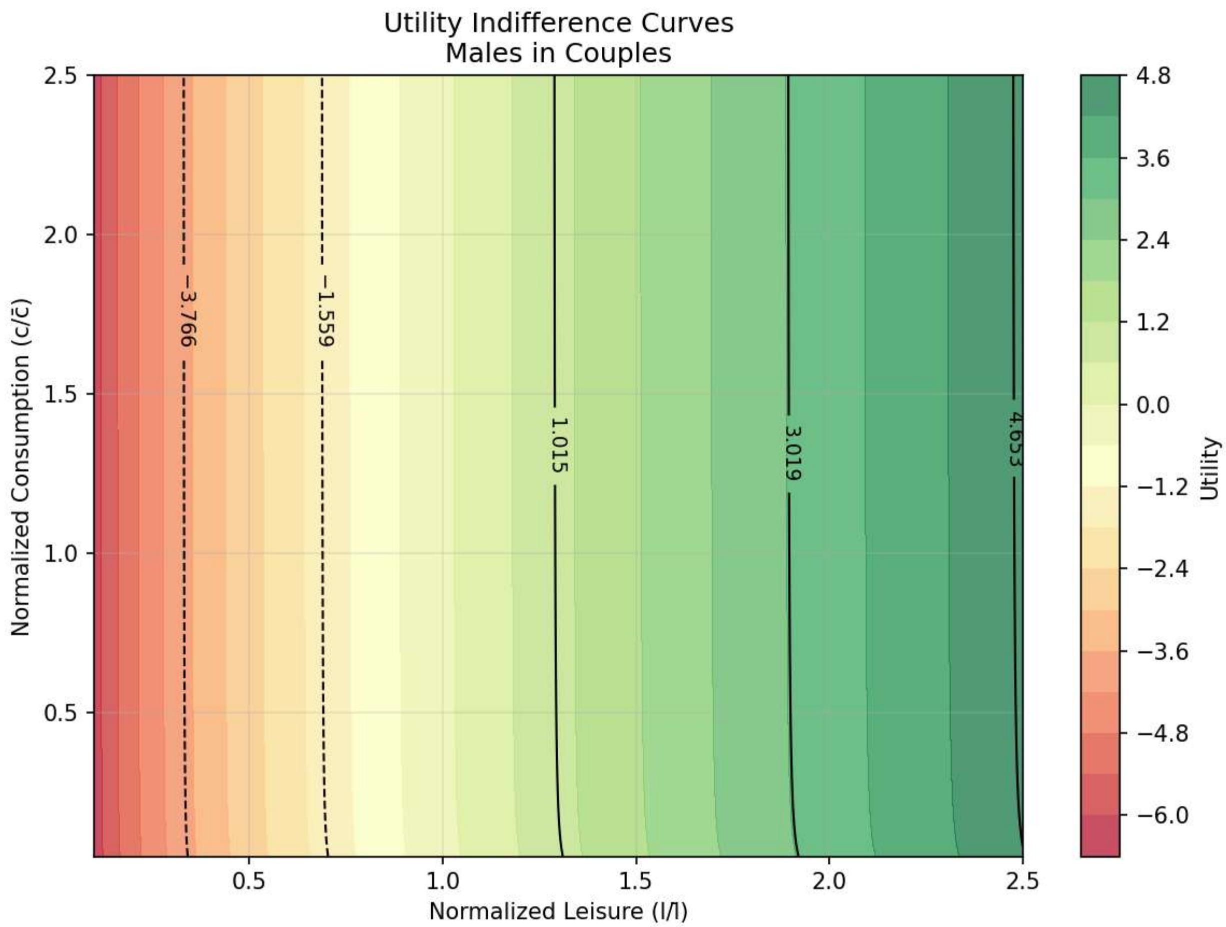
Single Males



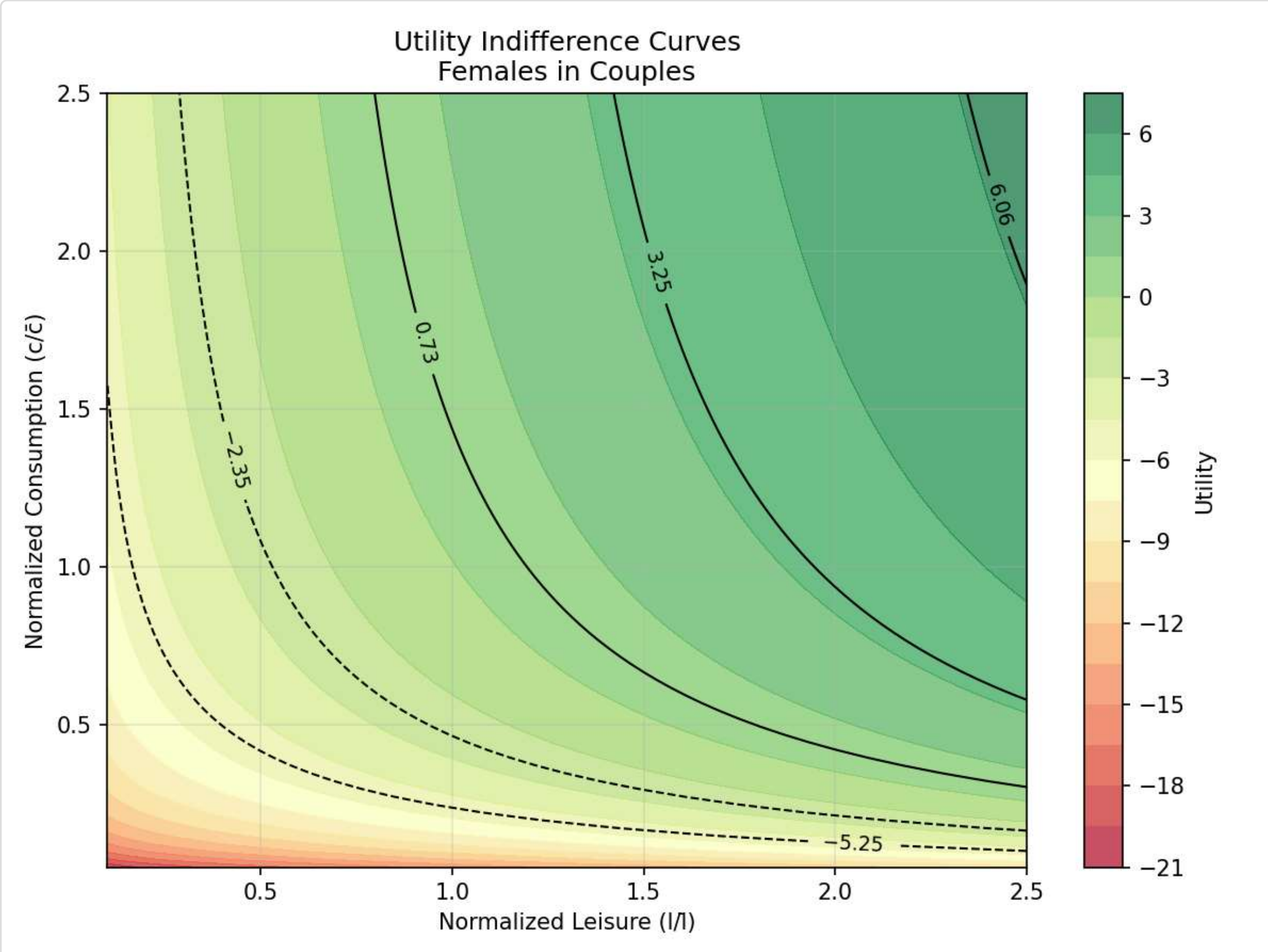
Single Females



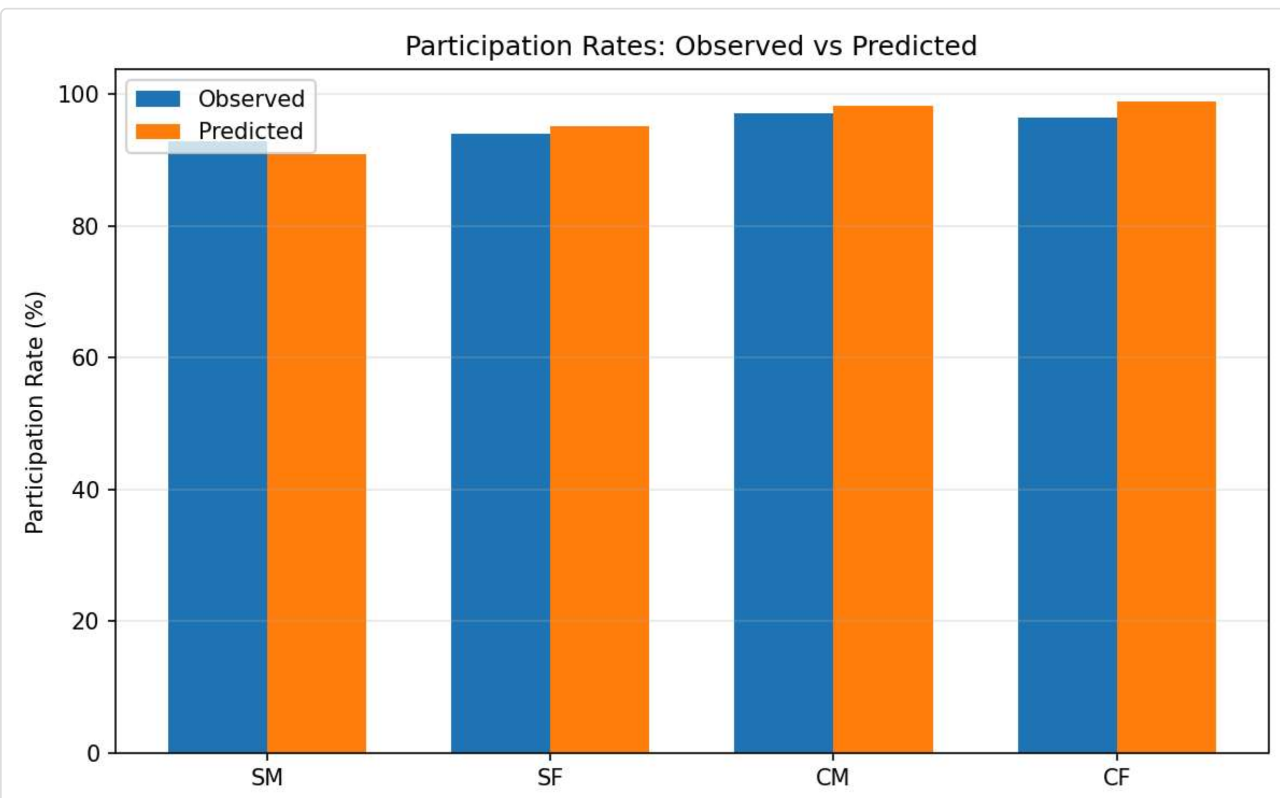
Males in Couples



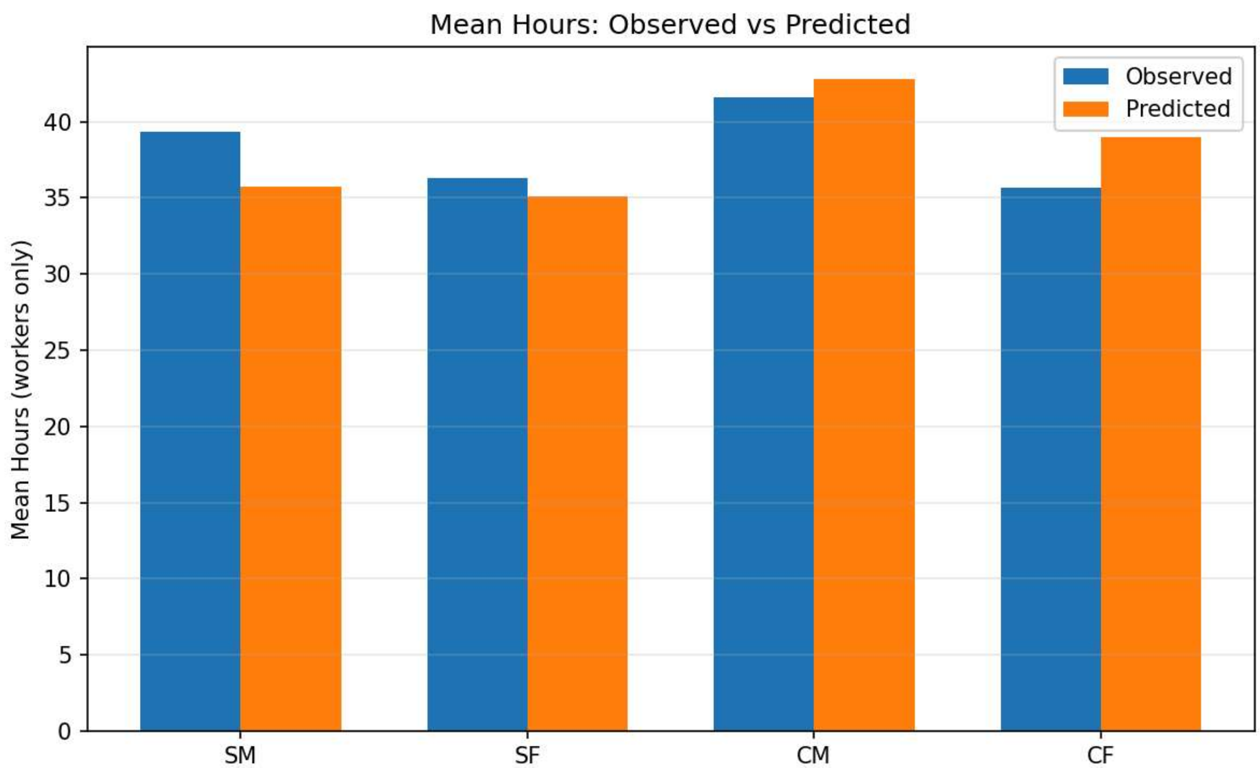
Females in Couples



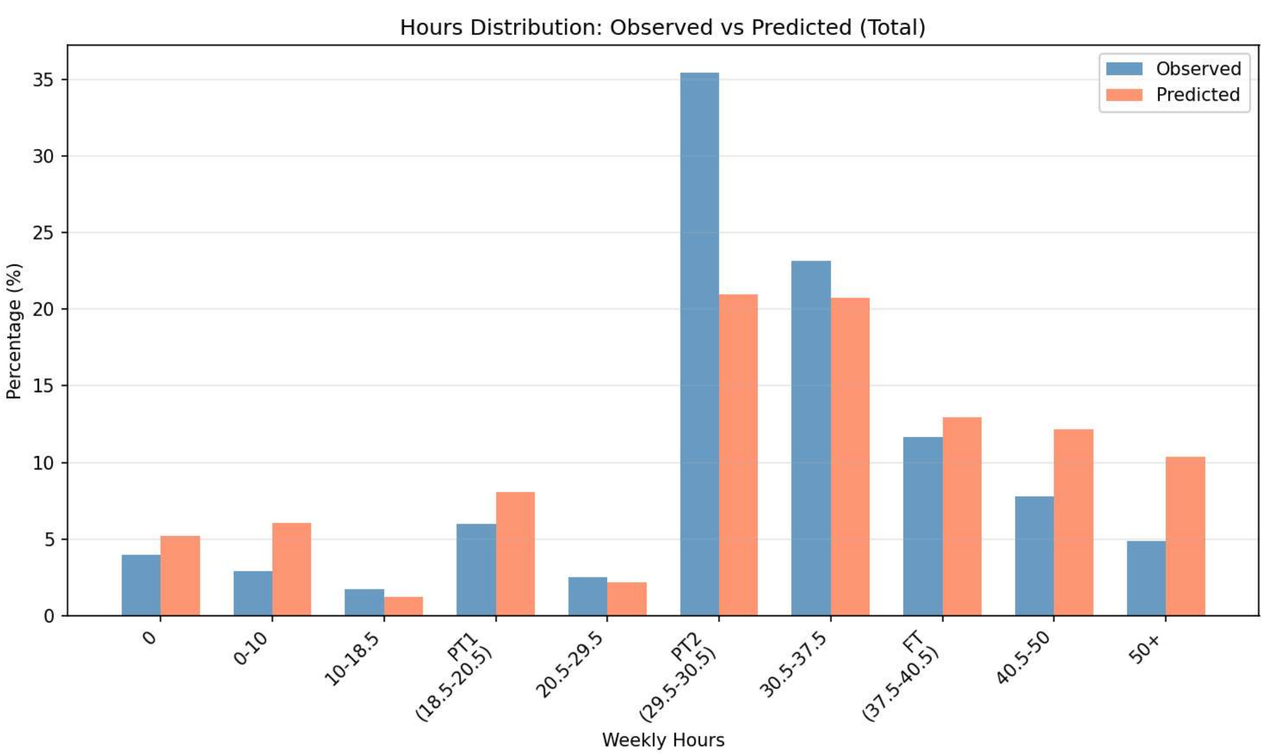
Participation

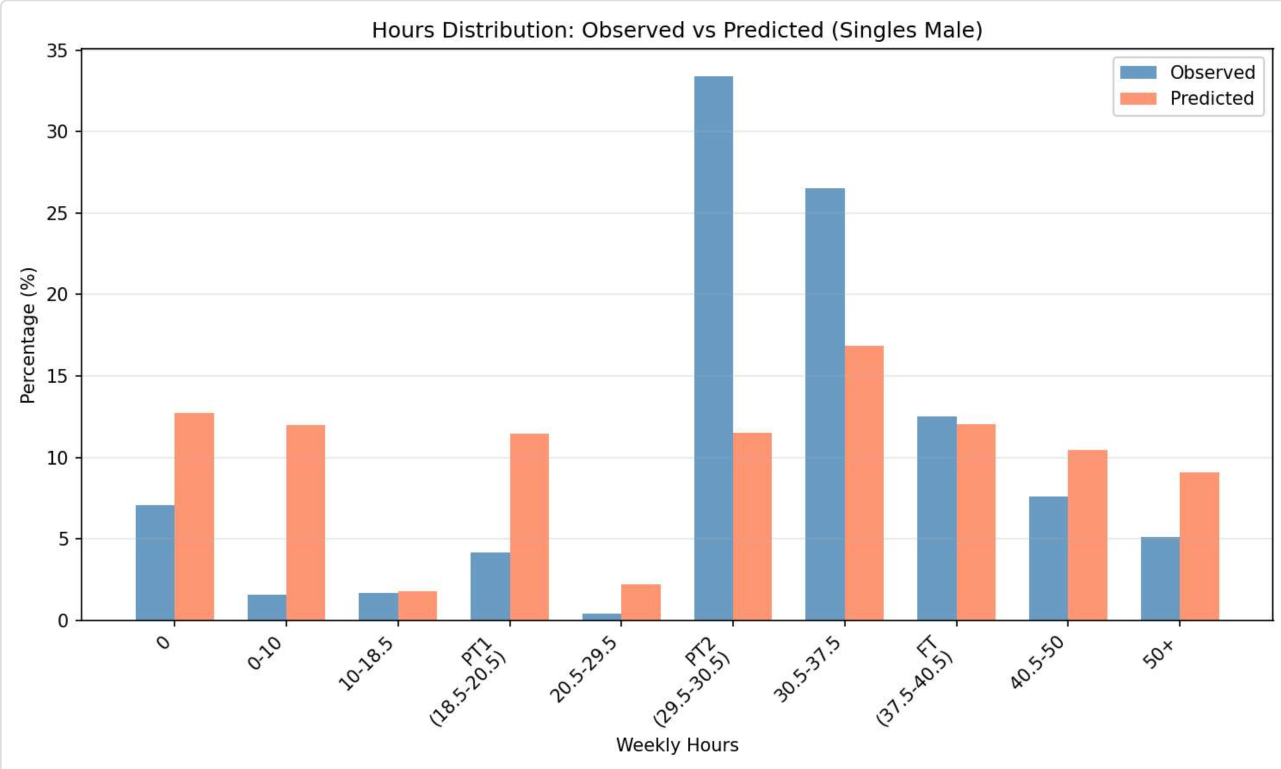


Mean Hours

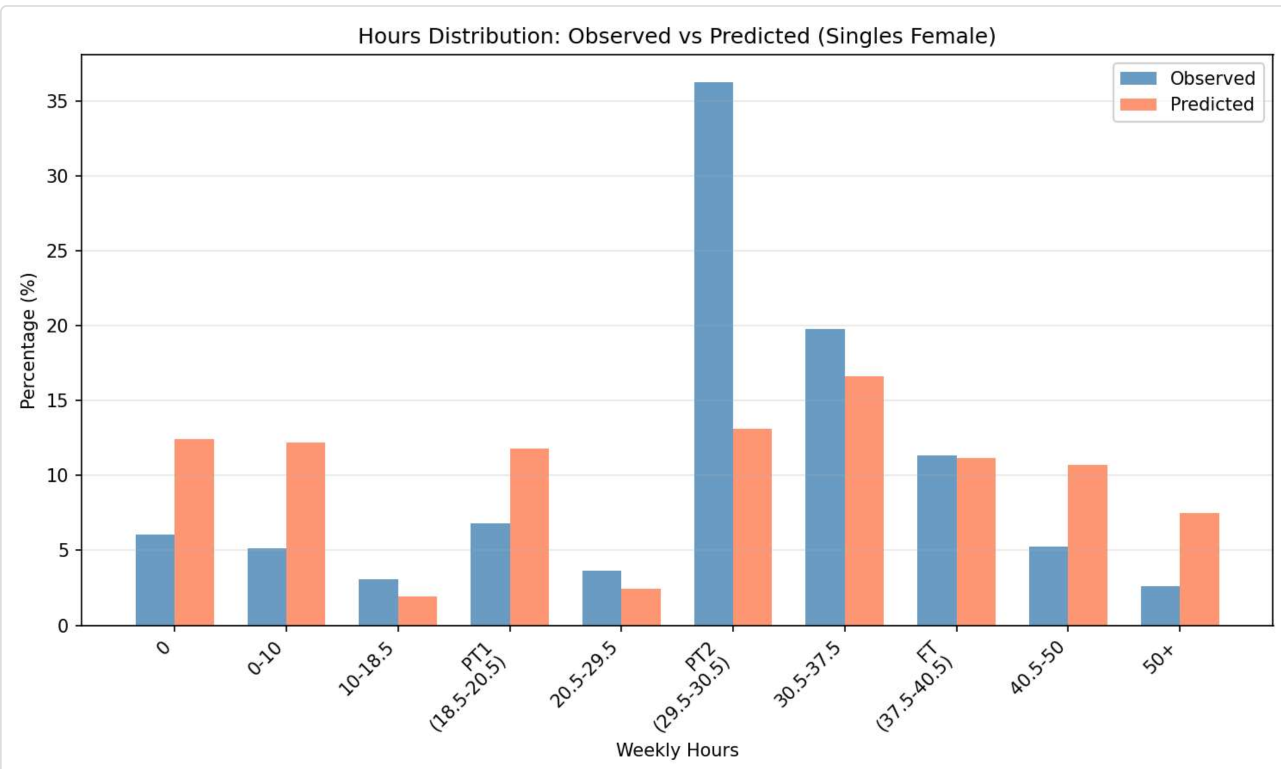


?? Hours Distribution: Observed vs Predicted

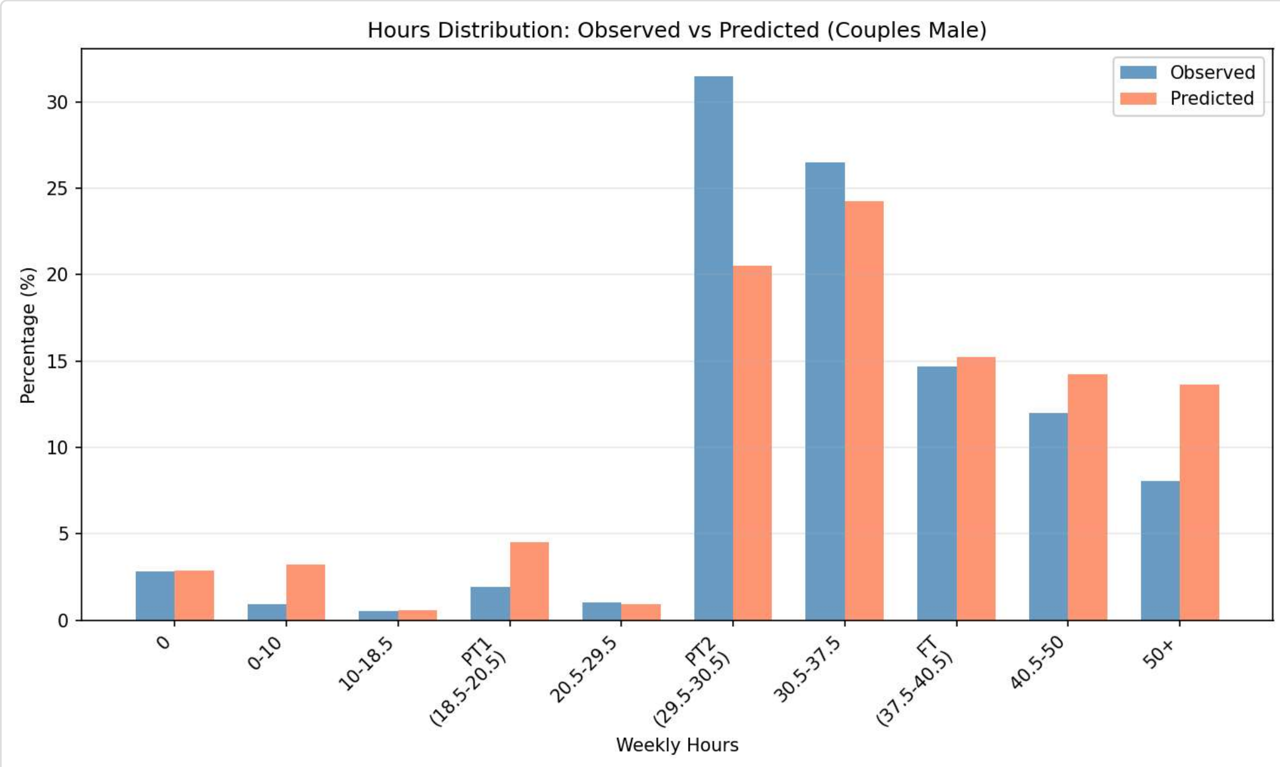


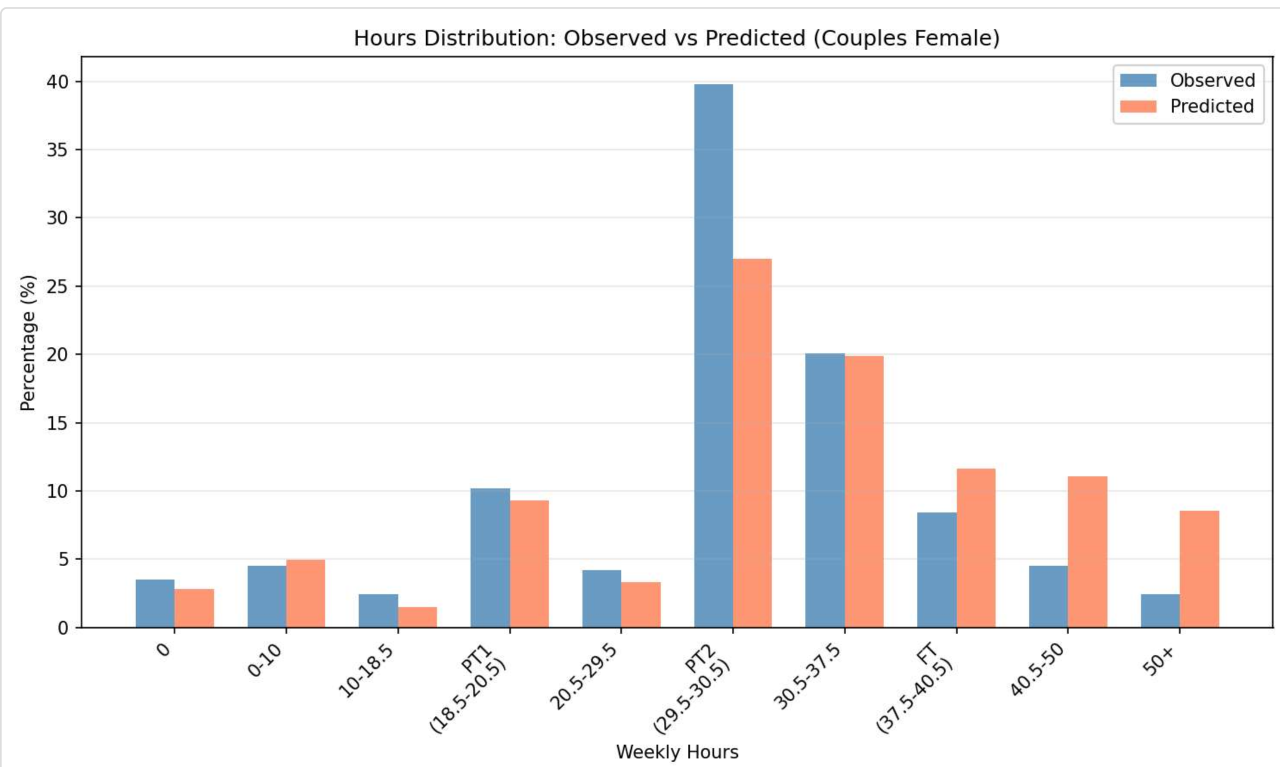


Hours Distribution: Observed vs Predicted (Singles Male)
Observed
Predicted
Percentage (%)
35
30
25
20
15
10
5
0
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+
Weekly Hours

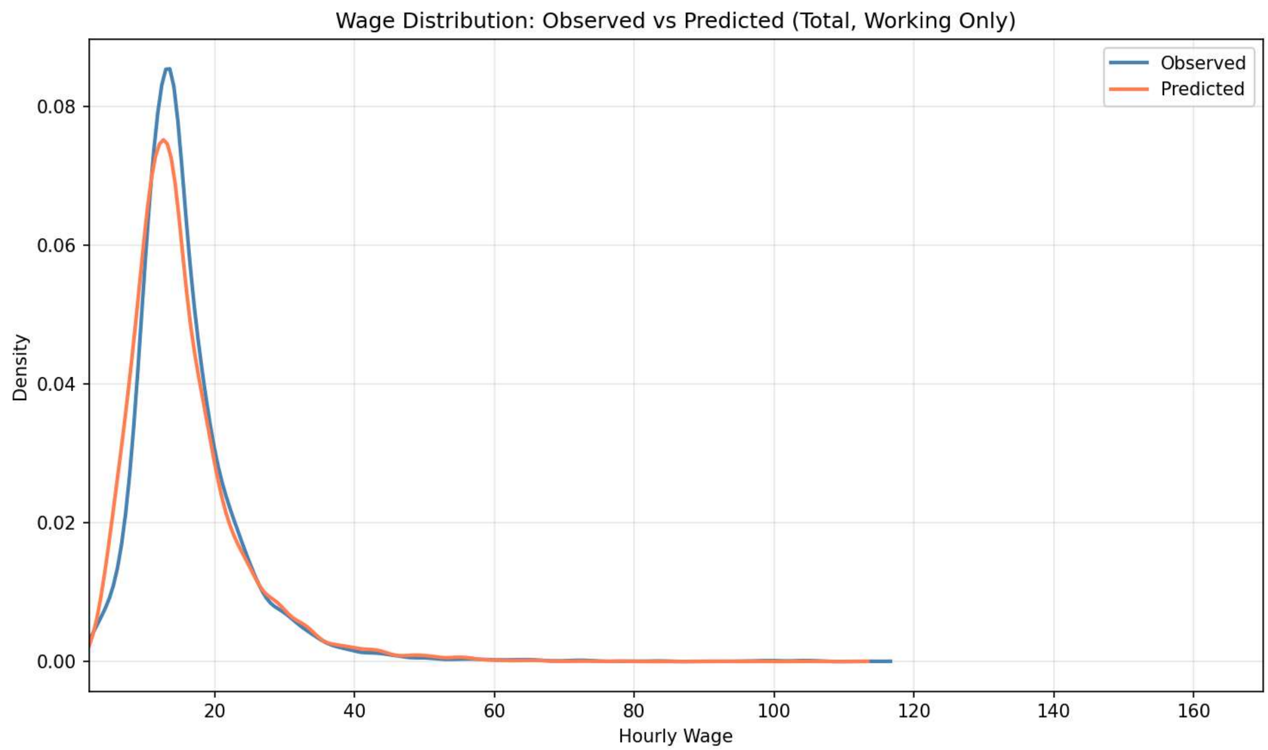


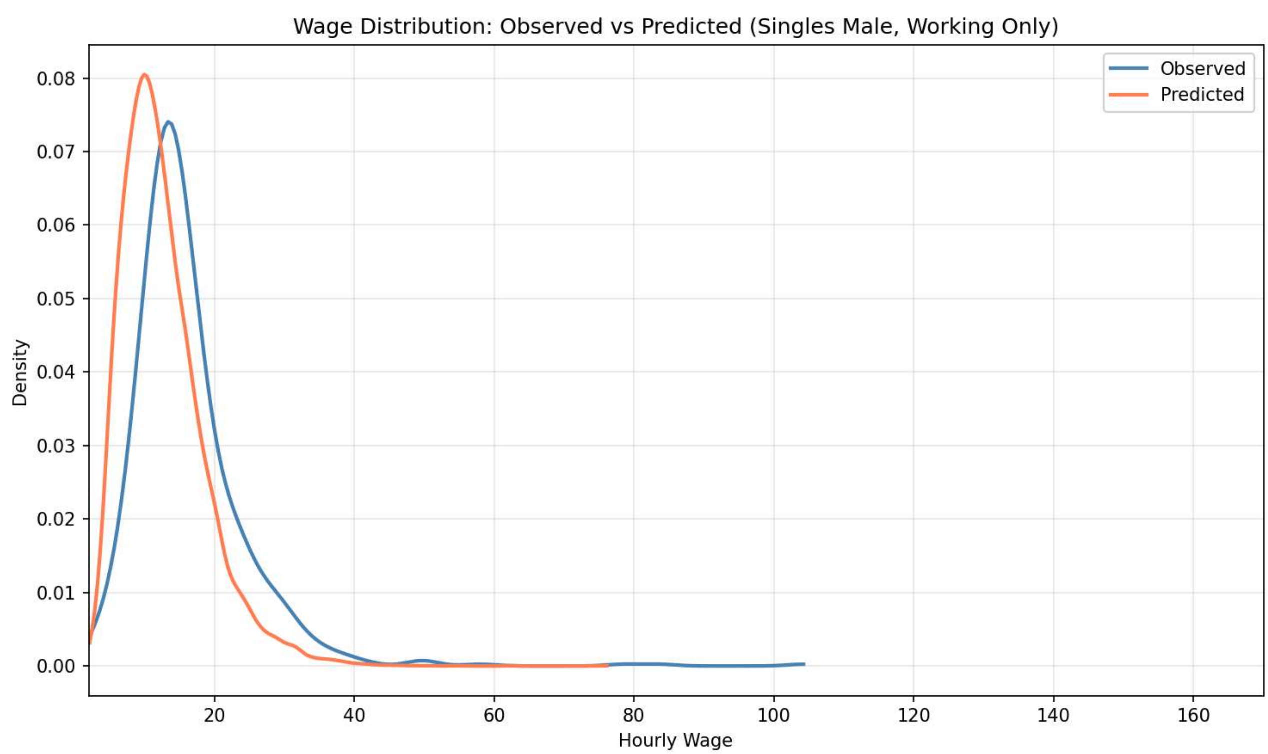
Hours Distribution: Observed vs Predicted (Singles Female)
Observed
Predicted
Percentage (%)
35
30
25
20
15
10
5
0
0
0-10
10-18.5
PT1 (18.5-20.5)
20.5-29.5
PT2 (29.5-30.5)
30.5-37.5
FT (37.5-40.5)
40.5-50
50+
Weekly Hours

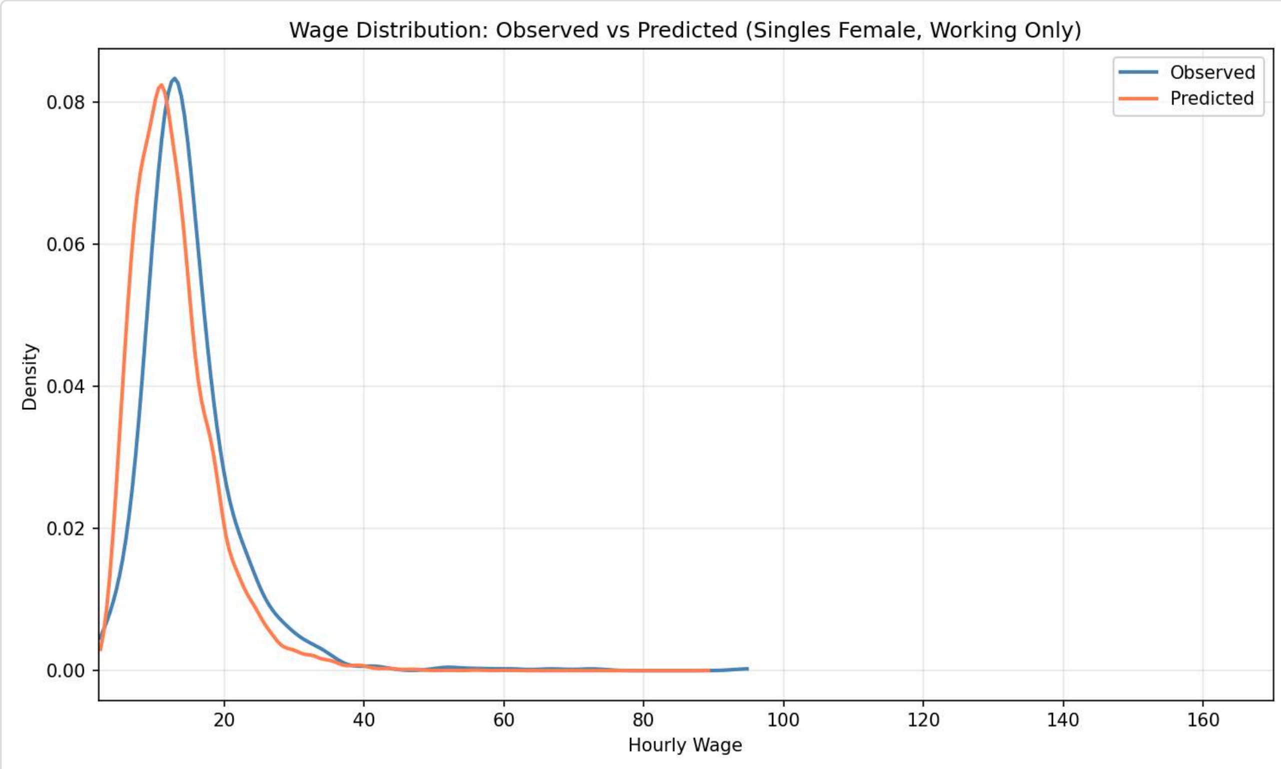


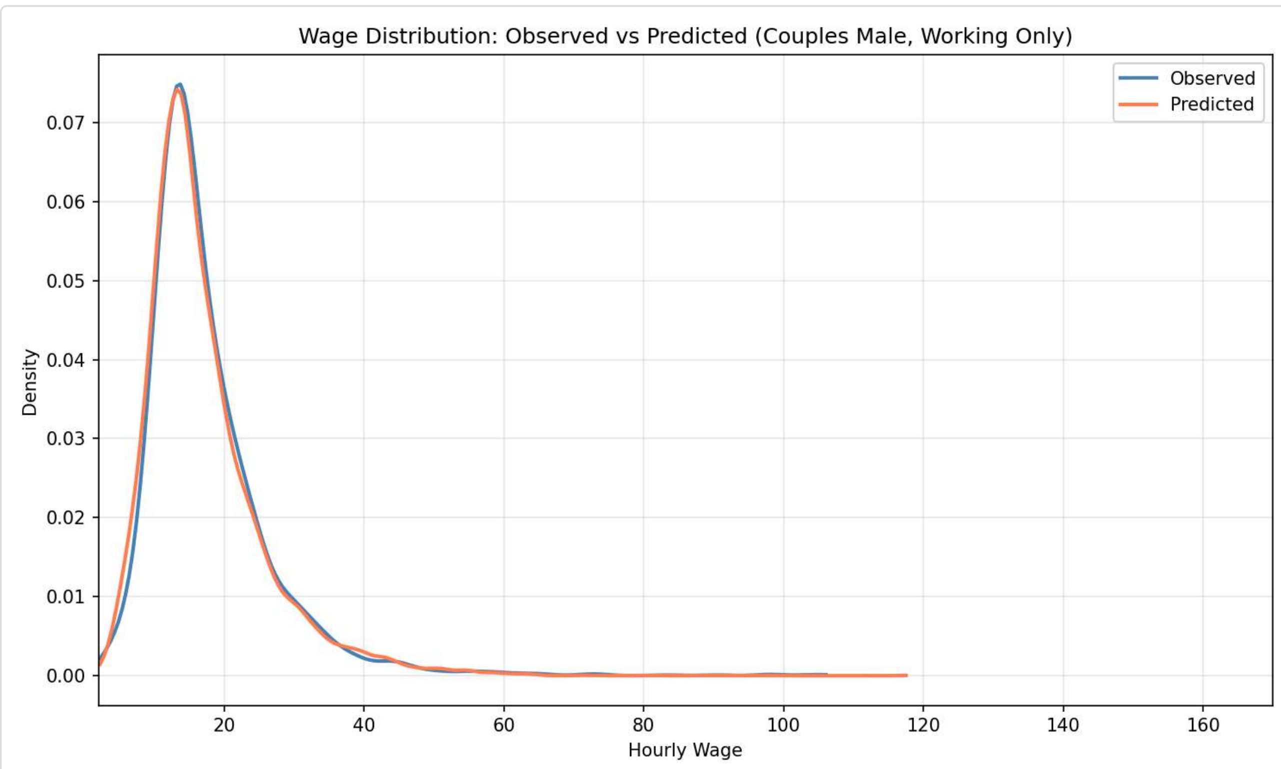


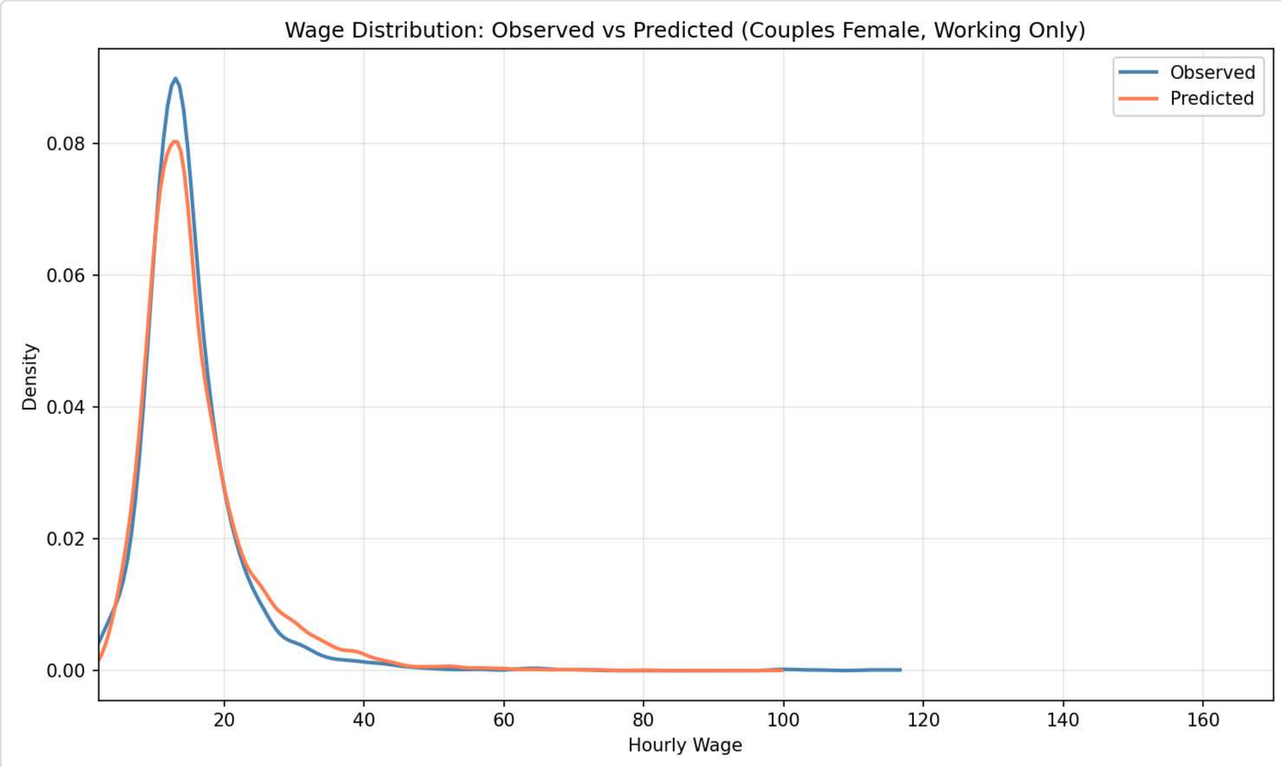
?? Wage Distribution: Observed vs Predicted (Working Only)


Wage Distribution: Observed vs Predicted (Total, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.02
0.04
0.06
0.08
20
40
60
80
100
120
140
160


Wage Distribution: Observed vs Predicted (Singles Male, Working Only)
Observed
Predicted
Density
Hourly Wage
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
20
40
60
80
100
120
140
160


Wage Distribution: Observed vs Predicted (Singles Female, Working Only)
Observed
Predicted
Density
0.00
0.02
0.04
0.06
0.08
20
40
60
80
100
120
140
160
Hourly Wage


Wage Distribution: Observed vs Predicted (Couples Male, Working Only)
Observed
Predicted
Density
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
20
40
60
80
100
120
140
160
Hourly Wage



Group-Specific Parameters

Single Males

beta_E	-2.8423
beta_E_educH	0.6134
beta_E_gsur	-0.7438
beta_c	0.6358
beta_h_ft	1.4441
beta_h_pt1	-0.4987
beta_h_pt2	0.3650
beta_l0	3.8740
beta_l_age	0.0085
beta_l_age2	0.0020
beta_ll	2.6237

Single Females

beta_E	-2.8423
beta_E_educH	0.6134
beta_E_gsur	-0.7438
beta_c	0.5760
beta_h_ft	1.4441
beta_h_pt1	-0.4987
beta_h_pt2	0.3650
beta_l0	4.4588
beta_l_age	0.0019
beta_l_age2	0.0041
beta_l_nkids	0.0567

beta_occ_2	-1.5104
beta_occ_2_cf	0.1763
beta_occ_2_cm	-1.4760
beta_occ_3	-2.1651
beta_occ_3_cf	-0.2164
beta_occ_3_cm	-2.2240
beta_occ_4	0.0237
beta_occ_4_cf	1.1147
beta_occ_4_cm	0.4725
beta_w0	2.0249
beta_w_educH	0.3161
beta_w_educL	-0.0510
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4268
theta_c_singles	-0.9357
theta_l	-0.7119

beta_ll	2.6237
beta_occ_2	-0.0105
beta_occ_2_cf	0.1763
beta_occ_2_cm	-1.4760
beta_occ_3	-0.5610
beta_occ_3_cf	-0.2164
beta_occ_3_cm	-2.2240
beta_occ_4	0.7988
beta_occ_4_cf	1.1147
beta_occ_4_cm	0.4725
beta_w0	2.0249
beta_w_educH	0.3161
beta_w_educL	-0.0510
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4268
theta_c_singles	-0.9357
theta_l	-0.7280

Males in Couples

beta_E	-2.8423
beta_E_educH	0.6134
beta_E_gsur	-0.7438
beta_c	4.0515
beta_h_ft	1.4441
beta_h_pt1	-0.4987
beta_h_pt2	0.3650
beta_l0	0.0119
beta_l_age	-0.0079
beta_l_age2	0.0006
beta_ll	2.6237

Females in Couples

beta_E	-2.8423
beta_E_educH	0.6134
beta_E_gsur	-0.7438
beta_c	4.0515
beta_h_ft	1.4441
beta_h_pt1	-0.4987
beta_h_pt2	0.3650
beta_l0	2.5821
beta_l_age	-0.0574
beta_l_age2	0.0028
beta_l_nkids	0.1769

beta_occ_2_cf	0.1763
beta_occ_2_cm	-1.4760
beta_occ_3_cf	-0.2164
beta_occ_3_cm	-2.2240
beta_occ_4_cf	1.1147
beta_occ_4_cm	0.4725
beta_w0	2.0249
beta_w_educH	0.3161
beta_w_educL	-0.0510
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4268
theta_c_singles	-0.9357
theta_l	-0.7319

beta_ll	2.6237
beta_occ_2_cf	0.1763
beta_occ_2_cm	-1.4760
beta_occ_3_cf	-0.2164
beta_occ_3_cm	-2.2240
beta_occ_4_cf	1.1147
beta_occ_4_cm	0.4725
beta_w0	2.0249
beta_w_educH	0.3161
beta_w_educL	-0.0510
beta_w_pexp	0.0181
beta_w_pexp2	-0.0002
sigma	0.4268
theta_c_singles	-0.9357
theta_l	-0.6777

📋 Parameter Estimates by Category

*Significance: *** p<0.001, ** p<0.01, * p<0.05*

☐ Bounded (has constraints) ☐ ⚠ Hit bound ☐ Warning: degenerate SE (near zero) ☐ p ∈ [0.05, 0.1) ☐ p ∈ [0.1, 0.25) ☐ p ≥ 0.25

🎯 Preference Parameters (Utility Function)

Parameters determining the utility from consumption (β_c, θ_c) and leisure (β_l, θ_l) by demographic group.*

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l0_sm	3.8740	0.7208	5.37	0.0000 ***	0.0500	50.0000	1.0000
joint.beta_l_age_sm	0.0085	0.0248	0.34	0.7323	-5.0000	5.0000	0.0000

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_l_age2_sm	0.0020	0.0021	0.97	0.3303	-1.0000	1.0000	0.0000
joint.beta_c_sm	0.6358	N/A	N/A	N/A	0.0500	50.0000	1.0000
joint.theta_l_sm	-0.7119	0.1546	-4.60	0.0000 ***	-8.0000	0.9500	-1.0000
joint.beta_l0_sf	4.4588	0.7807	5.71	0.0000 ***	0.0500	50.0000	1.0000

⏰ Employment and Hours Opportunity Parameters

Working-gated employment intercept and hours focal-point indicators, plus residual market-access shifters (gsur, education) declared in `market_opportunity.shifters`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_E	-2.8423	0.2991	-9.50	0.0000 ***	-25.0000	25.0000	0.0000
joint.beta_h_pt1	-0.4987	0.1079	-4.62	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_h_pt2	0.3650	0.1113	3.28	0.0010 **	-10.0000	10.0000	0.0000
joint.beta_h_ft	1.4441	0.0501	28.85	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_E_gsur	-0.7438	0.2207	-3.37	0.0008 ***	-10.0000	10.0000	0.0000
joint.beta_E_educH	0.6134	0.2361	2.60	0.0094 **	-10.0000	10.0000	0.0000

💰 Wage Opportunity / Mincer Parameters

Log-normal wage opportunity: $\mu_W(X)$ intercept, education, experience, and residual standard deviation σ.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_w0	2.0249	0.0252	80.24	0.0000 ***	-10.0000	20.0000	2.0000
joint.beta_w_educL	-0.0510	0.0211	-2.41	0.0158 *	-5.0000	5.0000	-0.1000
joint.beta_w_educH	0.3161	0.0150	21.14	0.0000 ***	-5.0000	5.0000	0.2000
joint.beta_w_pexp	0.0181	0.0022	8.11	0.0000 ***	-1.0000	1.0000	0.0200
joint.beta_w_pexp2	-0.0002	0.0000	-4.42	0.0000 ***	-0.1000	0.1000	-0.0003
joint.sigma	0.4268	0.0040	107.84	0.0000 ***	0.1000	20.0000	0.5000

🧭 Occupation Opportunity Parameters

Working-gated occupation shifters declared in `occupation_opportunity.shifters`; routed by `applies_to` group (sm/sf/cm/cf). Reference category is the YAML's `occupation_opportunity.reference`.

Parameter	Estimate	Std Error	t-value	p-value	Lower Bound	Upper Bound	Initial Value
joint.beta_occ_2_sm	-1.5104	0.1421	-10.63	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_sm	-2.1651	0.1842	-11.75	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sm	0.0237	0.0860	0.27	0.7833	-10.0000	10.0000	0.0000
joint.beta_occ_2_sf	-0.0105	0.1126	-0.09	0.9260	-10.0000	10.0000	0.0000
joint.beta_occ_3_sf	-0.5610	0.1290	-4.35	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_sf	0.7988	0.0920	8.68	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cm	-1.4760	0.1138	-12.97	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_3_cm	-2.2240	0.1480	-15.03	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_4_cm	0.4725	0.0688	6.87	0.0000 ***	-10.0000	10.0000	0.0000
joint.beta_occ_2_cf	0.1763	0.1003	1.76	0.0788	-10.0000	10.0000	0.0000
joint.beta_occ_3_cf	-0.2164	0.1117	-1.94	0.0526	-10.0000	10.0000	0.0000

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